

Omkar Singh

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SUMMARY

Full-stack developer and B.Tech student at IIT Kharagpur with hands-on experience building distributed systems in Go and production-grade web applications with the MERN stack and React.js. Designed backends processing 50–100 orders/sec at sub-ms latency. Seeking full-time SDE / full-stack roles to ship scalable, real-world software.

EDUCATION

Indian Institute of Technology (IIT) Kharagpur <i>B.Tech – Industrial & Systems Engineering</i> Coursework: Data Structures & Algorithms, Process Design, Project Management, Optimization	2023 – Present <i>GPA: 7.56 / 10.0</i>
Douglas Memorial Higher Secondary School, Barrackpore <i>ISC (Class 12)</i>	2021 – 2023 <i>80.6%</i>
Douglas Memorial Higher Secondary School, Barrackpore <i>ICSE (Class 10)</i>	2021 <i>85%</i>

EXPERIENCE

JPMorganChase – Software Engineering Job Simulation (Forage) <i>Software Engineering Virtual Experience</i> - Integrated Kafka into a Spring Boot microservice to consume and deserialize high-volume transaction messages using a configurable topic and embedded Kafka test framework, enabling reliable async event processing at scale. - Implemented transaction validation and persistence with Spring Data JPA and H2 SQL, modelling entities and balance updates across 3+ relational user-record tables. - Developed a REST endpoint for user-balance queries via a Spring controller; connected the service to an external Incentive API using RestTemplate, incorporating responses into end-to-end transactional workflows. - Verified system behaviour across 5+ test scenarios using Maven test suites and debugger-driven inspection, ensuring reliability across message ingestion, database operations, and external API interactions.	Apr – May 2026 <i>Remote</i>
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PROJECTS

Synthetic-Bull <i>Go, Gin, GORM, MySQL, WebSockets, REST APIs</i> Live - Architected an 11-node distributed Go backend handling real-time order matching, portfolio management, and market data aggregation across decoupled microservices at sub-ms inter-service latency. - Implemented a custom Limit Order Book with Price-Time Priority: $O(\log N)$ insertion via Red-Black Trees, $O(1)$ best-price retrieval, supporting partial fills across concurrent sessions. - Engineered lock-free fan-out via goroutines and sync.WaitGroup broadcasting to 5 services at sub-ms latency; exposed REST API Gateway + WebSocket endpoints for tick-by-tick candlestick rendering. - Built a GBM market simulator generating 50–100 realistic orders/sec, driving live OHLCV candlestick aggregation with real-time frontend updates.	2025
Collaborative Whiteboard <i>React.js, Firebase Firestore, Canvas API, Tailwind CSS</i> Live - Architected a multi-user drawing app supporting 10+ concurrent users with sub-100ms sync latency via Firebase Firestore real-time listeners propagating canvas state across all clients. - Built a stateful drawing engine with 6 tools and full undo/redo history via Canvas API and Rough.js; designed a mobile-first responsive UI with React Hooks.	Aug 2025
MERN E-Commerce Platform <i>MongoDB, Express.js, React.js, Node.js, JWT, Multer</i> Live - Built a full-stack e-commerce platform with JWT auth via HTTP-only cookies (XSS-safe), role-based admin controls, product image upload via Multer, and complete order + review management. - Designed a RESTful API with Express.js and Mongoose featuring async error-handling, bcrypt hashing, and protected route guards; deployed React SPA to Netlify with a concurrent dev workflow.	2025

TECHNICAL SKILLS

Languages	Java, Go, Python, JavaScript (ES6+), TypeScript, SQL
Frameworks & Libs	React.js, Node.js, Express.js, Spring Boot, Gin/GORM, Tailwind CSS
Databases	MongoDB, MySQL, Firebase Firestore, H2
Tools & Platforms	Git, Docker, REST APIs, WebSockets, JWT, Kafka, Netlify, Maven
Concepts	Microservices, Distributed Systems, DSA, Agile/Scrum, OOP

CERTIFICATIONS

Introduction to Software Engineering Job Simulation Commonwealth Bank (Forage)	May 2026
Software Engineering Job Simulation Quantum (Forage)	May 2026
Advanced Software Engineering Simulation Walmart Global Tech (Forage)	Sep 2025

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

LeetCode	Solved 250+ problems across arrays, trees, graphs, and dynamic programming
Codeforces	Rating 1286 (Specialist) – consistent performance in algorithmic contests
HackerRank	5-Star Gold Badge in Problem Solving